

Question 1)

```
mysql> CREATE DATABASE BankDB;
Query OK, 1 row affected (0.13 sec)

mysql> USE BankDB;
Database changed
mysql> CREATE TABLE Customer (
    ->     CustomerID INT PRIMARY KEY,
    ->     CustomerName VARCHAR(50),
    ->     Age INT,
    ->     City VARCHAR(30),
    ->     Balance DECIMAL(10,2)
    -> );
Query OK, 0 rows affected (0.18 sec)

mysql> DESC Customer;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| CustomerID     | int           | NO   | PRI | NULL    |       |
| CustomerName   | varchar(50)   | YES  |     | NULL    |       |
| Age            | int           | YES  |     | NULL    |       |
| City           | varchar(30)   | YES  |     | NULL    |       |
| Balance        | decimal(10,2) | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.06 sec)

mysql> ALTER TABLE Customer
    -> ADD Phone VARCHAR(15);
Query OK, 0 rows affected (0.12 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE Customer
    -> ADD Phone VARCHAR(15);
ERROR 1060 (42S21): Duplicate column name 'Phone'
mysql> ALTER TABLE Customer
```

```
mysql> ALTER TABLE Customer
    -> ADD Phone VARCHAR(15);
Query OK, 0 rows affected (0.12 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE Customer
    -> ADD Phone VARCHAR(15);
ERROR 1060 (42S21): Duplicate column name 'Phone'
mysql> ALTER TABLE Customer
    -> MODIFY CustomerName VARCHAR(100);
Query OK, 0 rows affected (0.11 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE Customer
    -> RENAME COLUMN City TO Address;
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE Customer
    -> RENAME COLUMN City TO Address;
ERROR 1054 (42S22): Unknown column 'City' in 'customer'
mysql> ALTER TABLE Customer
    -> DROP COLUMN Phone;
Query OK, 0 rows affected (0.19 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> RENAME TABLE Customer TO Customers;
Query OK, 0 rows affected (0.04 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_bankdb |
+-----+
| customers         |
+-----+
1 row in set (0.01 sec)
```

```
mysql> TRUNCATE TABLE Customers;
Query OK, 0 rows affected (0.06 sec)

mysql> SELECT * FROM Customers;
Empty set (0.00 sec)

mysql> DROP TABLE Customers;
Query OK, 0 rows affected (0.04 sec)

mysql> SHOW TABLES;
Empty set (0.00 sec)

mysql> CREATE DATABASE BankDB;
ERROR 1007 (HY000): Can't create database 'bankdb'; database exists
mysql>
mysql> USE BankDB;
Database changed
mysql>
mysql> CREATE TABLE Customer(
->     CustomerID INT PRIMARY KEY,
->     CustomerName VARCHAR(50),
->     Age INT,
->     City VARCHAR(30),
->     Balance DECIMAL(10,2)
-> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql>
mysql> DESC Customer;
```

Field	Type	Null	Key	Default	Extra
CustomerID	int	NO	PRI	NULL	
CustomerName	varchar(50)	YES		NULL	
Age	int	YES		NULL	
City	varchar(30)	YES		NULL	
Balance	decimal(10,2)	YES		NULL	

```
5 rows in set (0.00 sec)
```

```
mysql>
mysql> ALTER TABLE Customer
    -> ADD Phone VARCHAR(15);
Query OK, 0 rows affected (0.08 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> ALTER TABLE Customer
    -> MODIFY CustomerName VARCHAR(100);
Query OK, 0 rows affected (0.09 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> ALTER TABLE Customer
    -> RENAME COLUMN City TO Address;
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> ALTER TABLE Customer
    -> DROP COLUMN Phone;
Query OK, 0 rows affected (0.08 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> RENAME TABLE Customer TO Customers;
Query OK, 0 rows affected (0.03 sec)
```

```
mysql>
mysql> TRUNCATE TABLE Customers;
Query OK, 0 rows affected (0.05 sec)
```

```
mysql>
mysql> DROP TABLE Customers;
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> |
```

Q2)

```
mysql> USE Bank_Management;
Database changed
mysql> CREATE TABLE bank_transactions (
->     txn_id INT PRIMARY KEY,
->     customer_name VARCHAR(50),
->     branch_name VARCHAR(50),
->     transaction_type VARCHAR(20),
->     amount DECIMAL(10,2),
->     transaction_date DATE
-> );
```

Query OK, 0 rows affected (0.06 sec)

```
mysql> DESC bank_transactions;
```

Field	Type	Null	Key	Default	Extra
txn_id	int	NO	PRI	NULL	
customer_name	varchar(50)	YES		NULL	
branch_name	varchar(50)	YES		NULL	
transaction_type	varchar(20)	YES		NULL	
amount	decimal(10,2)	YES		NULL	
transaction_date	date	YES		NULL	

6 rows in set (0.02 sec)

```
mysql> DESC bank_transactions;
```

Field	Type	Null	Key	Default	Extra
txn_id	int	NO	PRI	NULL	
customer_name	varchar(50)	YES		NULL	
branch_name	varchar(50)	YES		NULL	
transaction_type	varchar(20)	YES		NULL	
amount	decimal(10,2)	YES		NULL	
transaction_date	date	YES		NULL	

```

+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> ALTER TABLE bank_transactions
    -> ADD account_no VARCHAR(20);
Query OK, 0 rows affected (0.08 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> DESC bank_transactions;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| txn_id         | int           | NO   | PRI | NULL    |       |
| customer_name  | varchar(50)   | YES  |     | NULL    |       |
| branch_name    | varchar(50)   | YES  |     | NULL    |       |
| transaction_type | varchar(20)   | YES  |     | NULL    |       |
| amount         | decimal(10,2) | YES  |     | NULL    |       |
| transaction_date | date          | YES  |     | NULL    |       |
| account_no     | varchar(20)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql> ALTER TABLE bank_transactions
    -> MODIFY customer_name VARCHAR(100);
Query OK, 0 rows affected (0.09 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> RENAME TABLE bank_transactions
    -> TO customer_transactions;
Query OK, 0 rows affected (0.04 sec)

mysql> RENAME TABLE customer_transactions
    -> TO bank_transactions;
Query OK, 0 rows affected (0.04 sec)

mysql> TRUNCATE TABLE bank_transactions;
Query OK, 0 rows affected (0.05 sec)

```

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```
mysql> TRUNCATE TABLE bank_transactions;
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE bank_backup (
  ->   id INT
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO bank_transactions
  -> (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
  -> VALUES
  -> (101, 'Ravi', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
  -> (102, 'Sita', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
  -> (103, 'Kiran', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
  -> (104, 'Anil', 'Vizag', 'Deposit', 8000, '2024-01-10'),
  -> (105, 'Priya', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
  -> (106, 'Ramesh', 'Vizag', 'Deposit', 15000, '2024-01-12'),
  -> (107, 'Keerthi', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
  -> (108, 'Rahul', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
  -> (109, 'Sneha', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
  -> (110, 'Madhu', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
Query OK, 10 rows affected (0.04 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> INSERT INTO bank_transactions
  -> (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
  -> VALUES
  -> (111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18');
Query OK, 1 row affected (0.02 sec)

mysql> UPDATE bank_transactions
  -> SET amount=5000
  -> WHERE txn_id=105;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
-> WHERE txn_id=111;
Query OK, 1 row affected (0.02 sec)

mysql> SELECT * FROM bank_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 5000.00 | 2024-01-05 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 | 2024-01-06 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 12000.00 | 2024-01-08 | NULL |
| 104 | Anil | Vizag | Deposit | 8000.00 | 2024-01-10 | NULL |
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | NULL |
| 106 | Ramesh | Vizag | Deposit | 15000.00 | 2024-01-12 | NULL |
| 107 | Keerthi | Vijayawada | Withdrawal | 1000.00 | 2024-01-13 | NULL |
| 108 | Rahul | Hyderabad | Deposit | 9000.00 | 2024-01-14 | NULL |
| 109 | Sneha | Vizag | Withdrawal | 4000.00 | 2024-01-15 | NULL |
| 110 | Madhu | Vijayawada | Deposit | 11000.00 | 2024-01-16 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
10 rows in set (0.04 sec)

mysql> SELECT *
  -> FROM bank_transactions
  -> WHERE transaction_type='Deposit';
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 5000.00 | 2024-01-05 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 12000.00 | 2024-01-08 | NULL |
| 104 | Anil | Vizag | Deposit | 8000.00 | 2024-01-10 | NULL |
| 106 | Ramesh | Vizag | Deposit | 15000.00 | 2024-01-12 | NULL |
| 108 | Rahul | Hyderabad | Deposit | 9000.00 | 2024-01-14 | NULL |
| 110 | Madhu | Vijayawada | Deposit | 11000.00 | 2024-01-16 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
6 rows in set (0.02 sec)

mysql> SELECT *
  -> FROM bank_transactions
```

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```
mysql> SELECT *
-> FROM bank_transactions
-> ORDER BY amount DESC;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 106 | Ramesh | Vizag | Deposit | 15000.00 | 2024-01-12 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 12000.00 | 2024-01-08 | NULL |
| 110 | Madhu | Vijayawada | Deposit | 11000.00 | 2024-01-16 | NULL |
| 108 | Rahul | Hyderabad | Deposit | 9000.00 | 2024-01-14 | NULL |
| 104 | Anil | Vizag | Deposit | 8000.00 | 2024-01-10 | NULL |
| 101 | Ravi | Hyderabad | Deposit | 5000.00 | 2024-01-05 | NULL |
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | NULL |
| 109 | Sneha | Vizag | Withdrawal | 4000.00 | 2024-01-15 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 | 2024-01-06 | NULL |
| 107 | Keerthi | Vijayawada | Withdrawal | 1000.00 | 2024-01-13 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> CREATE USER 'Auditor1'@'localhost'
-> IDENTIFIED BY 'audit123';
Query OK, 0 rows affected (0.10 sec)

mysql> CREATE USER 'BranchManager'@'localhost'
-> IDENTIFIED BY 'manager123';
Query OK, 0 rows affected (0.02 sec)

mysql> GRANT ALL PRIVILEGES
-> ON Bank_Management.bank_transactions
-> TO 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.02 sec)

mysql> REVOKE SELECT
-> ON Bank_Management.bank_transactions
-> FROM 'Auditor1'@'localhost';
ERROR 1147 (42000): There is no such grant defined for user 'Auditor1' on host 'localhost' on table 'bank_transactions'

mysql> REVOKE ALL PRIVILEGES
-> ON Bank_Management.bank_transactions
-> FROM 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> SET autocommit = 0;
Query OK, 0 rows affected (0.02 sec)

mysql> UPDATE bank_transactions
-> SET amount=6000
-> WHERE txn_id=101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> COMMIT;
Query OK, 0 rows affected (0.02 sec)

mysql> SAVEPOINT Before_Update;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE bank_transactions
-> SET amount=99999
-> WHERE txn_id=102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM bank_transactions
-> WHERE txn_id=102;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 102 | Sita | Hyderabad | Withdrawal | 99999.00 | 2024-01-06 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> ROLLBACK TO Before_Update;
Query OK, 0 rows affected (0.02 sec)
```


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```
mysql> ROLLBACK TO Before_Update;  
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> SELECT * FROM bank_transactions  
-> WHERE txn_id=102;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL

```
1 row in set (0.00 sec)
```

```
mysql> UPDATE bank_transactions  
-> SET amount=7000  
-> WHERE txn_id=101;  
Query OK, 1 row affected (0.00 sec)  
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql>  
mysql> SAVEPOINT SP1;  
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>  
mysql> UPDATE bank_transactions  
-> SET amount=9000  
-> WHERE txn_id=102;  
Query OK, 1 row affected (0.00 sec)  
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql>  
mysql> ROLLBACK TO SP1;  
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>  
mysql> COMMIT;  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql>  
mysql> ROLLBACK TO SP1;  
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>  
mysql> COMMIT;  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM bank_transactions  
-> WHERE txn_id IN (101,102);
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	7000.00	2024-01-05	NULL
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL

```
2 rows in set (0.00 sec)
```

```
mysql> |
```